

ABM BUSINESS LOGIC BIBLE

§5 v2 (DEEP) — TIER ASSIGNMENT & ROUTING

Second-Order Excavation · 14 Reviewer Questions · 12 Resolved · 2 Flagged Open

§5 (the operational summary) specified the thresholds, hysteresis, fresh-signal gate, engagement block, and routing actions. This deeper pass excavates the second-order questions an engineering reviewer would press on — the places where the original spec is operationally underdetermined and two developers could implement different behavior and both believe they followed the rules. Twelve are resolved as additional rules. Two are **genuinely undecided** and are flagged as **OPEN DESIGN QUESTIONS** rather than fabricated. ‘Don’t know’ is a first-class spec state — it tells engineering where decisions still need to be made.

Reviewer questions	14	Resolved as rules	12
Open design questions	2 (flagged)	New rules	17

THE OPEN-DESIGN-QUESTION PATTERN (NEW)

This document introduces a discipline the Bible has not used before: **OPEN-DESIGN-QUESTION** as a first-class artifact. Flagged questions carry: the question, candidate options, trade-off each implies, and the rule IDs that depend on resolution. They must be resolved before that area ships. This is more honest than guessing a default — a fabricated rule looks complete but pretends to a decision that hasn’t been made, which is exactly the EPIS-RCM-04 false-confidence failure mode applied to the spec itself.

A. THRESHOLDS & HYSTERESIS — SECOND-ORDER QUESTIONS

Q1. What happens if a manual override and hysteresis-exit collide?

A human force-sets the account to HOT (override). Two days later the score drops to 65 — below the 70 hysteresis-exit. Does the override hold, or does the engine demote? §5 said overrides expire in 30 days but didn't say what blocks them sooner.

Resolution: **override is authoritative for its full 30-day duration**. Score-based exit thresholds do not fire while a manual override is active; only the override's natural expiry or an explicit override-removal action ends it. This matches GOV: humans outrank the engine on the same decision.

Q2. "Score crosses 78" — spike trigger or stability trigger?

A score jumping from 60 to 82 in a single recalc and a score sitting at 78 for ten recalcs are very different. Does either trigger HOT?

Resolution: **crossing is a single-recalc event**. The hysteresis band itself absorbs noise (entry 78, exit 70 = 8-point dead zone); requiring N consecutive recalcs would add latency without adding stability beyond what hysteresis already provides. A 60→82 jump is a P1 signal landing — the engine must respond in hours, not by waiting for confirmation recalcs.

Q3. Boundary semantics — inclusive or exclusive at 78 / 70 / 53 / 47?

V2 used '≥75' (inclusive) and 'drops below 70' (exclusive). The deep pass must pick one for each boundary or the bands aren't cleanly defined.

Resolution: **entries are inclusive, exits are strict**. Enter HOT when score ≥ 78; exit HOT when score ≤ 70. Enter WARM when score ≥ 53; exit WARM (downward) when score ≤ 47. The asymmetry reinforces stickiness: it is easier to stay in a tier than to leave it.

B. OWNER ASSIGNMENT & ROTATION — SECOND-ORDER QUESTIONS

Q4. Round-robin discipline on edge cases (PTO, load, skill) — OPEN

Owner on PTO, owner already has 8 HOT accounts, owner is the only one with KSA banking domain expertise. V2 said 'rotation' but never said which kind. You answered 'don't know' — the right answer.

[OPEN-DESIGN-QUESTION OPEN-Q4]

Question: What discipline governs owner assignment on HOT promotion?

Candidate options: (a) Pure round-robin — simple, fair, ignores PTO / load / skill. (b) Round-robin with PTO + load skip — simple + practical; skips owners on PTO or above a HOT-count threshold. (c) Weighted by KSA banking expertise — best fit, slower onboarding for new SDRs, requires expertise registry.

Trade-off: (a) is fastest to ship and easiest to audit but produces real friction. (b) is the operationally honest middle. (c) requires an expertise model the engine doesn't yet have.

Dependent rules: ROUTE-02 (owner-assignment routing), TIER-UP-02 (assign owner on HOT promotion). These rules will be specified once OPEN-Q4 is resolved; until then engineering must implement only the (a) pure round-robin as the safe default, behind a feature flag for upgrade to (b) or (c).

Q5. What happens to the owner on demotion that doesn't trigger removal?

HOT → WARM with active outreach 'let it complete' — owner stays. But what about HOT → WARM with no active outreach, or WARM → COLD?

Resolution: **owner is released on tier-down when no active outreach exists.** Owner is released immediately on HOT→WARM with no in-flight sequence; on WARM→COLD owner is always released. The released slot is returned to the rotation pool. If the account is re-promoted later it is treated as a fresh assignment.

Q6. Self-override vs admin-override — who counts as the human authority?

The assigned owner re-tiers their own account. Is that an admin override (logged like one) or a routine sales action?

Resolution: **self-override by the assigned owner is logged as an override but does not require an admin role.** Owner-override carries a separate flag (owner_override = TRUE) for telemetry; it still triggers the 30-day expiry. Override by a non-owner non-admin is rejected.

C. DEMOTION DURING IN-FLIGHT OUTREACH — SECOND-ORDER QUESTIONS

Q7. What does “let active outreach complete” actually mean?

All four email touches over 14 days regardless of intermediate signal? Or stop at the next natural sequence checkpoint if engagement is absent?

Resolution: **complete means run to natural sequence end OR to the next negative-signal checkpoint**, whichever first. The sequence completes if engagement signals remain neutral; it auto-halts at the next checkpoint if cumulative engagement is below the active-engagement threshold (zero opens after Touch 2, for example). This avoids both premature termination and wasted touches into a dead account.

Q8. Race condition — reply arrives during the let-it-complete window after demotion. Does the account un-demote?

Account demoted HOT→WARM; sequence is completing; contact replies positively on Day 3 of the completion window. The reply triggers engagement-block but the account is already WARM.

Resolution: **positive engagement during the completion window auto-re-promotes the account to HOT** and re-engages the engagement-block, preventing further demotion. The original demotion is logged as preceded-by-engagement and reversed. This guarantees the engine never pulls back from a warming account.

D. FRESH-SIGNAL GATE — SECOND-ORDER QUESTIONS

Q9. Signal arrival time or signal event date for freshness?

A press release reaches the engine today covering an event that occurred 30 days ago. Is the signal FRESH (because we just received it) or STALE (because the event is old)?

Resolution: **freshness is anchored to event_date, not detected_at**. The engine cares about how recent the underlying event is, not how recently the engine learned of it. A signal's detected_at is preserved for audit but does not control freshness. EPIS-HALF-01 semantic half-life is also event-date-anchored.

Q10. Does a REGULATORY signal's fan-out create FRESH state on dormant accounts?

SAMA issues an open-banking directive. The fan-out (V2 SIG-DET-011) attaches the signal to every account in the affected segment, including dormant ones. Does a dormant account suddenly satisfy the fresh-signal gate purely from fan-out?

Resolution: **yes — a fanned-out high-trust regulatory signal satisfies the fresh-signal gate on every account it touches**. The whole point of fan-out is that the regulation creates uniform pressure across the segment; gating on prior account activity would defeat that. However, the regulatory dimension alone is unlikely to push composite above 78 without other dimensions; in practice this allows promotion only where reachability and warmth are already non-zero.

E. EDGE PREEMPTION & ROUTING — SECOND-ORDER QUESTIONS

Q11. Race condition — EDGE scenario activates between score-recalc and routing-action. Does routing fire?

Tier promotion fires at $t=0$; EDGE-02 (regulator action against the target) activates at $t=+50\text{ms}$ before the routing action executes. Does the routing fire then get rolled back, or does EDGE preempt?

Resolution: **EDGE is checked at routing-action commit time, not at recalc time.** The routing action atomically checks for an active EDGE scenario as its final precondition; if found, the routing is preempted rather than executed-then-rolled-back. This avoids the partial-state problem of rollback.

Q12. On EDGE clearance — replay the preempted routing, or re-resolve from current state?

An EDGE-02 (regulator action) clears 3 weeks after preempting a HOT promotion. Does the engine replay the original promotion, or re-score and re-route from today's state?

Resolution: **re-resolve from current state.** Three weeks of signal decay, new engagement, and possible new events mean the preempted promotion may no longer be the correct action. Replay would act on stale evidence — the same failure mode the fresh-signal gate exists to prevent. Re-resolution is the only honest behavior.

F. SCORE→TIER→ENRICHMENT→SIGNAL FEEDBACK LOOP — SECOND-ORDER QUESTIONS

Q13. Oscillation prevention in the feedback loop — OPEN

HOT promotion triggers enrichment; enrichment can generate new signals (HIRING_SURGE detection, V2 §6 Step 4); new signals trigger re-scoring; re-scoring may re-tier. Without a damper, this can oscillate. You answered 'don't know' — the right answer.

[OPEN-DESIGN-QUESTION OPEN-Q13]

Question: What prevents oscillation in the score→tier→enrichment→signal feedback loop?

Candidate options: (a) Tier-change cooldown — no further tier change for 24h after a promotion. Simple, easy to audit, may delay legitimate demotions. (b) Lockout — HOT promotion locks tier for the duration of enrichment + first sequence (≈3–10 days). Strongest stability, may keep clearly-wrong promotions stuck. (c) No cooldown — rely solely on hysteresis to dampen. Lowest latency, highest oscillation risk in feedback-rich periods.

Trade-off: hysteresis alone is the simplest design but doesn't protect against the specific enrichment-generates-signal feedback path (it dampens score noise, not derived-signal noise). A 24h cooldown is the empirically common choice in similar systems. A lockout is the most defensible against oscillation but couples tier to workflow stage.

Dependent rules: ROUTE-01 (atomic routing), TIER-UP-02 (HOT promotion side effects), and the enrichment-as-signal-generator path in §6 Step 4. Engineering must implement a minimum 24h tier-change cooldown as a safe default behind a feature flag pending resolution — this is the lowest-risk stance until OPEN-Q13 is decided.

Q14. What if enrichment fails on HOT promotion — does the tier roll back?

V2 says ENRICHING→WATCHING after 3 enrichment failures. But tier (HOT) and account_state (ENRICHING) are different fields. Does tier follow state back to a lower value?

Resolution: tier holds; account_state reverts. The account remains HOT (the scoring evidence is unchanged by the engine's inability to enrich) but account_state transitions ENRICHING→HOT_UNENRICHED, a new substate. EXEC stage 6 cannot run from HOT_UNENRICHED — outreach is blocked pending human review or retry. This makes the failure observable (the account stays visible as HOT in dashboards) without acting on it.

G. NEW RULES FROM THIS DEEP PASS

Rule ID	Specification
TIER-DEEP-01	Manual overrides are authoritative for their full 30-day duration; score-based exit thresholds do not fire while an override is active. Override removal or expiry hands authority back to scoring (resolves Q1).
TIER-DEEP-02	Tier crossing is a single-recalc event; the hysteresis band absorbs noise. A spike (e.g. 60→82) triggers HOT immediately. No multi-recalc confirmation is required (resolves Q2).
TIER-DEEP-03	Boundary semantics: enter HOT when score ≥ 78 ; exit HOT when score ≤ 70 ; enter WARM when ≥ 53 ; exit WARM (downward) when ≤ 47 . Entries inclusive, exits strict (resolves Q3).
TIER-DEEP-04	Owner is released on tier-down when no active outreach exists. HOT→WARM with no in-flight sequence releases owner immediately; WARM→COLD always releases. Released slots return to the rotation pool. Re-promotion is a fresh assignment (resolves Q5).
TIER-DEEP-05	Self-override by the assigned owner is logged with owner_override = TRUE and does not require an admin role; still triggers the 30-day override expiry. Override by a non-owner non-admin is rejected (resolves Q6).
TIER-DEEP-06	'Active outreach completes' (TIER-DOWN-01) means: the sequence runs to natural end OR auto-halts at the next checkpoint if cumulative engagement is below the active-engagement threshold (zero opens by Touch 2). No further capital spent into a dead account (resolves Q7).
TIER-DEEP-07	Positive engagement during the post-demotion completion window auto-re-promotes the account to HOT and re-engages the engagement-block. Original demotion is logged as preceded-by-engagement and reversed (resolves Q8).
TIER-DEEP-08	Signal freshness is anchored to event_date, not detected_at. Old events surfaced through new reporting do not satisfy the fresh-signal gate. detected_at is preserved for audit only (resolves Q9).
TIER-DEEP-09	A fanned-out high-trust regulatory signal (SAMA / CMA / Tadawul) satisfies the fresh-signal gate on every account in the affected segment, including dormant ones. Composite-score promotion remains conditional on the other dimensions (resolves Q10).
TIER-DEEP-10	EDGE scenarios are checked at routing-action commit time as the final precondition; if found, routing is preempted, not executed-then-rolled-back. No partial-state on preemption (resolves Q11).
TIER-DEEP-11	On EDGE clearance, the preempted routing is NOT replayed; the engine re-scores and re-routes from current state. Replay on stale evidence is forbidden (resolves Q12).
TIER-DEEP-12	Enrichment failure on HOT promotion: tier holds at HOT; account_state transitions ENRICHING → HOT_UNENRICHED. EXEC stage 6 is blocked from HOT_UNENRICHED. Failure is observable, not silent (resolves Q14).
TIER-OPEN-Q4	[OPEN-DESIGN-QUESTION] Owner-assignment discipline on HOT promotion. Engineering default behind feature flag: pure round-robin. Upgrades to PTO/load-aware or skill-weighted require resolution of OPEN-Q4.
TIER-OPEN-Q13	[OPEN-DESIGN-QUESTION] Oscillation prevention in the score→tier→enrichment→signal loop. Engineering default behind feature flag: 24h tier-change cooldown after a promotion. Lockout or no-cooldown variants require resolution of OPEN-Q13.
TIER-OPEN-DISCIPLINE-01	OPEN-DESIGN-QUESTION rules carry the [OPEN-DESIGN-QUESTION Q-ID] marker, a candidate-options list, a trade-off note, a dependent-rules list, and an engineering safe-default. They are first-class rules; they MUST be resolved before that area ships to production.
TIER-OPEN-DISCIPLINE-02	Resolution of an OPEN-DESIGN-QUESTION is an A4 governance action (GOV-MOD-01 family): a human owner decides, the spec is updated with a new rule replacing the OPEN, and the feature flag is removed.
TIER-OPEN-DISCIPLINE-03	OPEN-DESIGN-QUESTION rules are tracked in the V2 Placement Map under 'unresolved spec decisions'; the map MUST not show a section as 'Complete' if it contains an unresolved OPEN.

STATUS UPDATE — §5 IS NOT YET 'COMPLETE'

Per TIER-OPEN-DISCIPLINE-03, §5 contains two unresolved OPEN-DESIGN-QUESTIONS (OPEN-Q4 and OPEN-Q13). Its Placement Map status drops from **Complete** to **Complete-with-Opens** until they are resolved. Engineering may build against §5 using the documented safe defaults (pure round-robin owner assignment + 24h tier-change cooldown), with both behind feature flags so they can be upgraded once OPEN-Q4 / OPEN-Q13 land.